

WILDFIRE SMOKE AREA

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Acknowledgements: Professor Jun Wu, Professor Qingqing He

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INTRODUCTION

Project Overview:

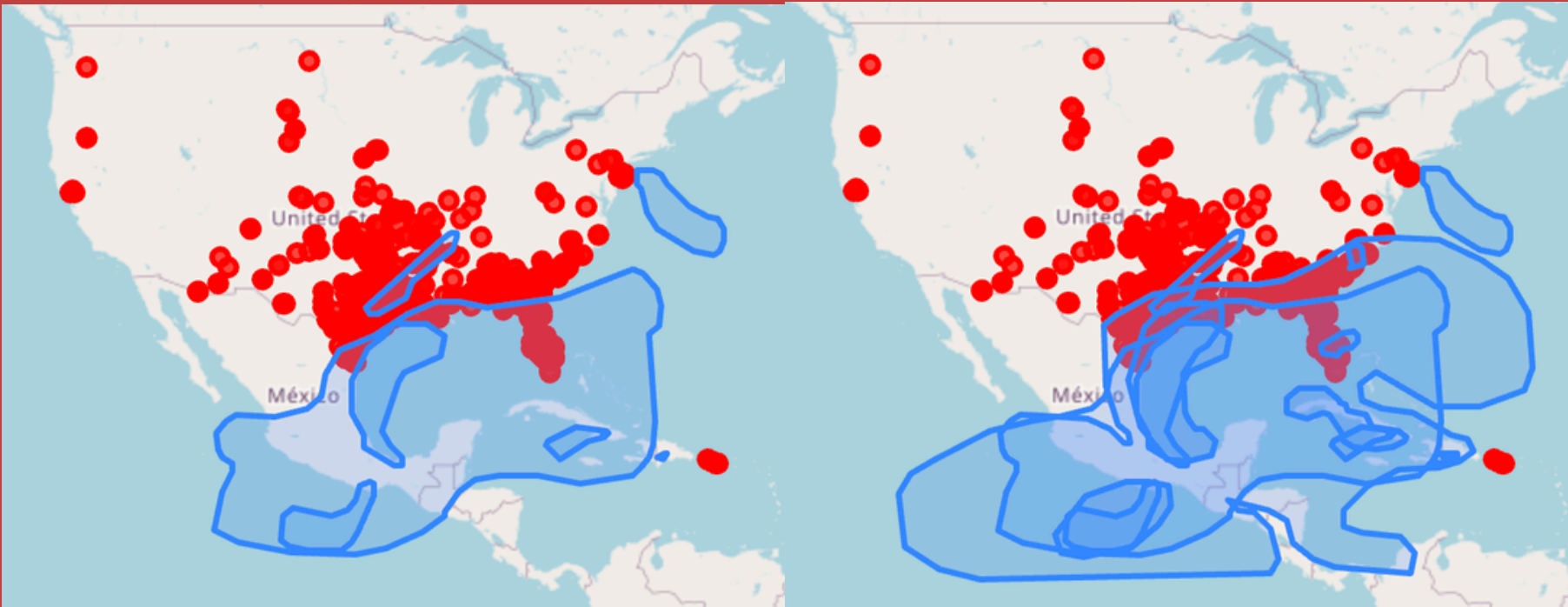
To analyze the spread of wildfire smoke as influenced by wildfire size, intensity, and weather conditions such as wind, temperature, and humidity.

End Goal

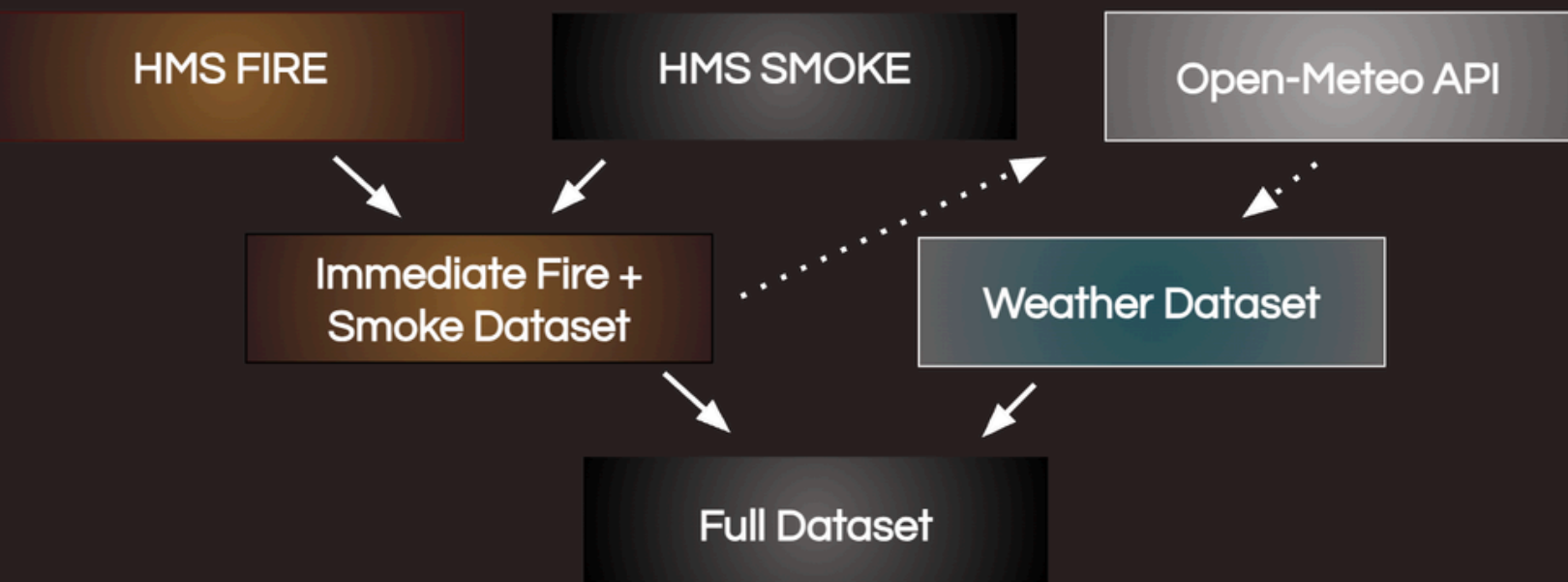
To create reliable methods for predicting how far and wide wildfire smoke will travel, helping communities better prepare for air quality impacts.

Data Sources:

NOAA Hazard Mapping System
Open-Meteo Historical Weather API

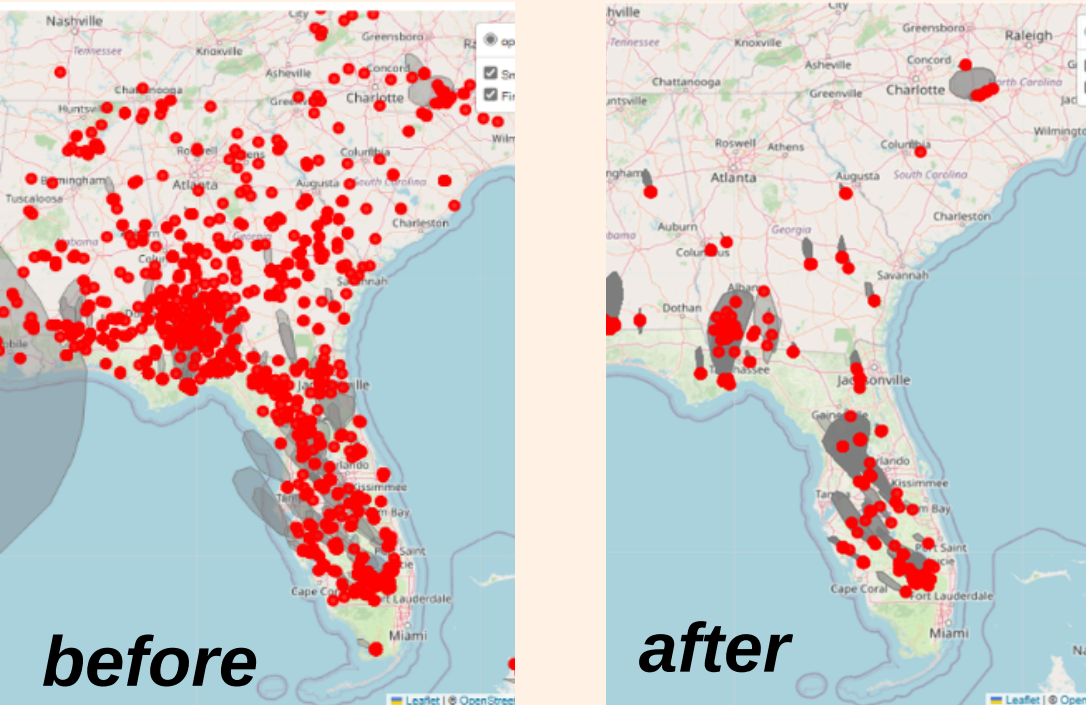


Database Workflow

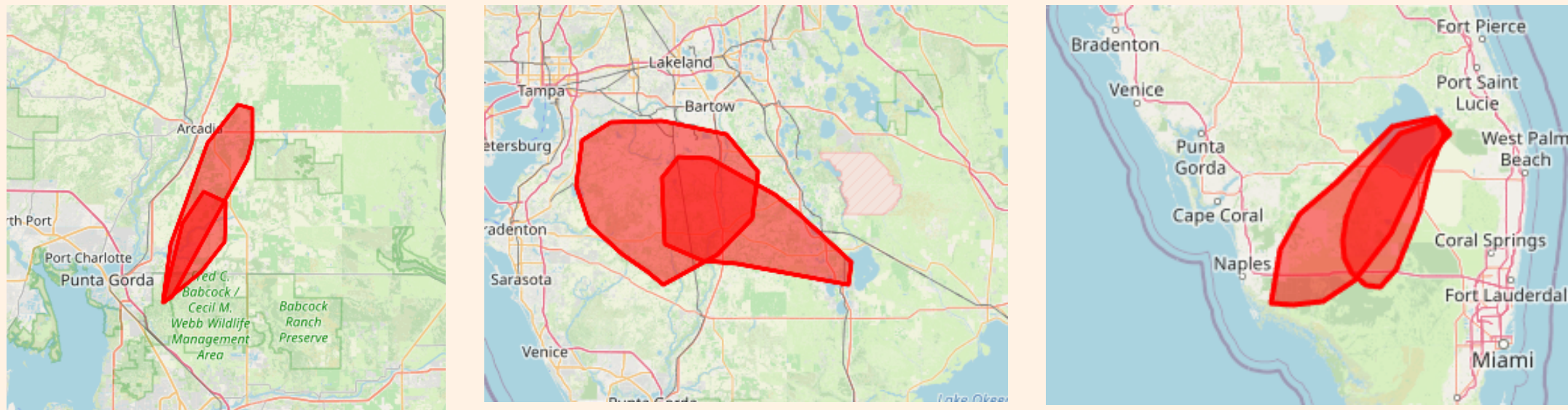


METHODOLOGY & ANALYSIS

Geospatial Techniques:

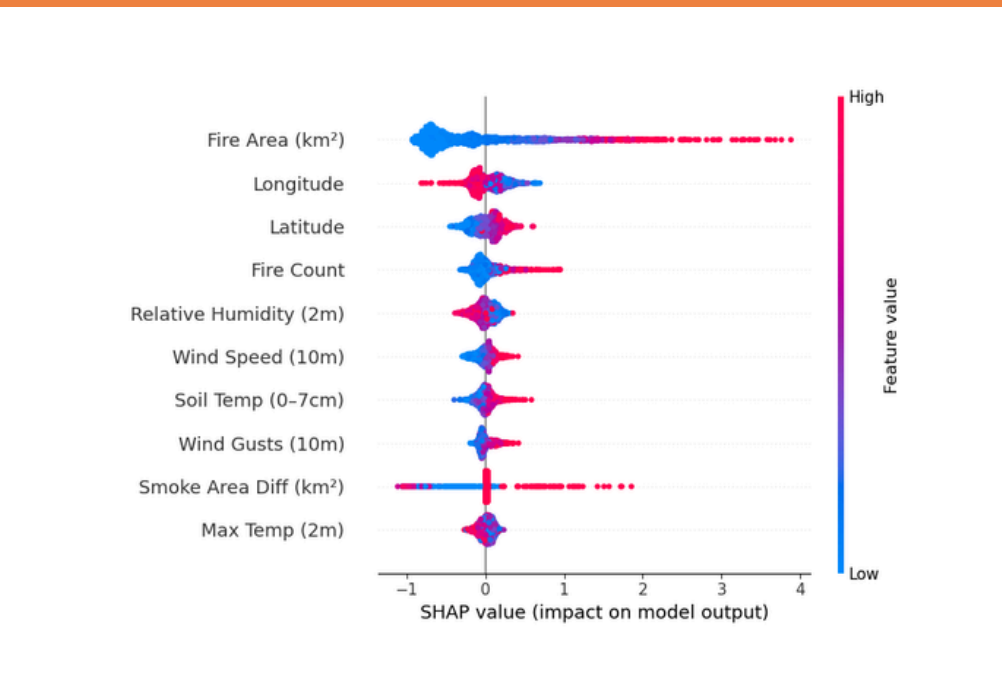
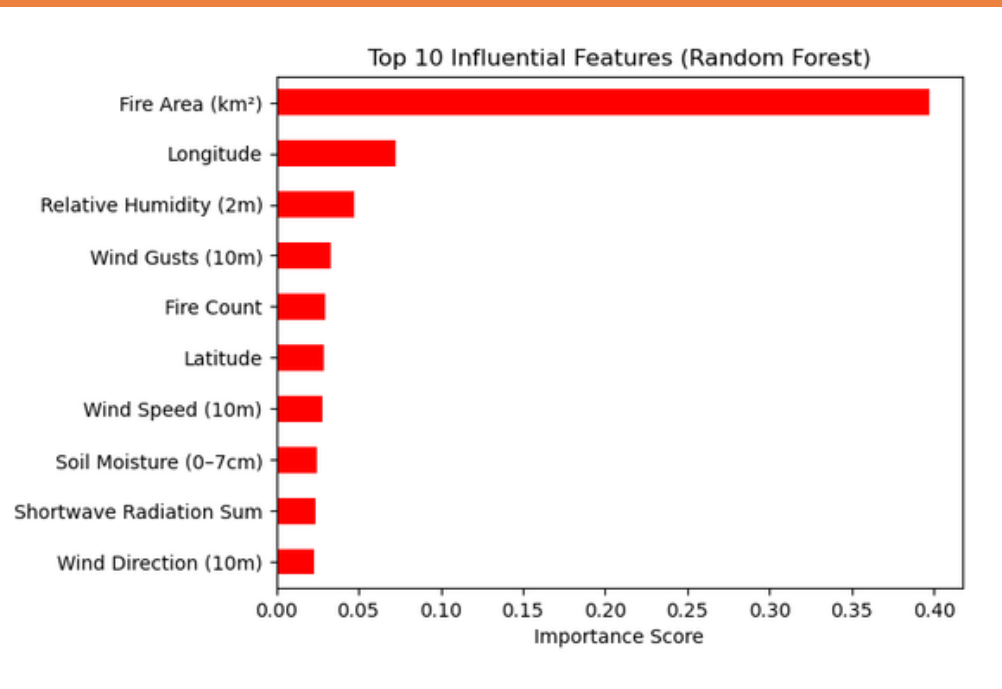
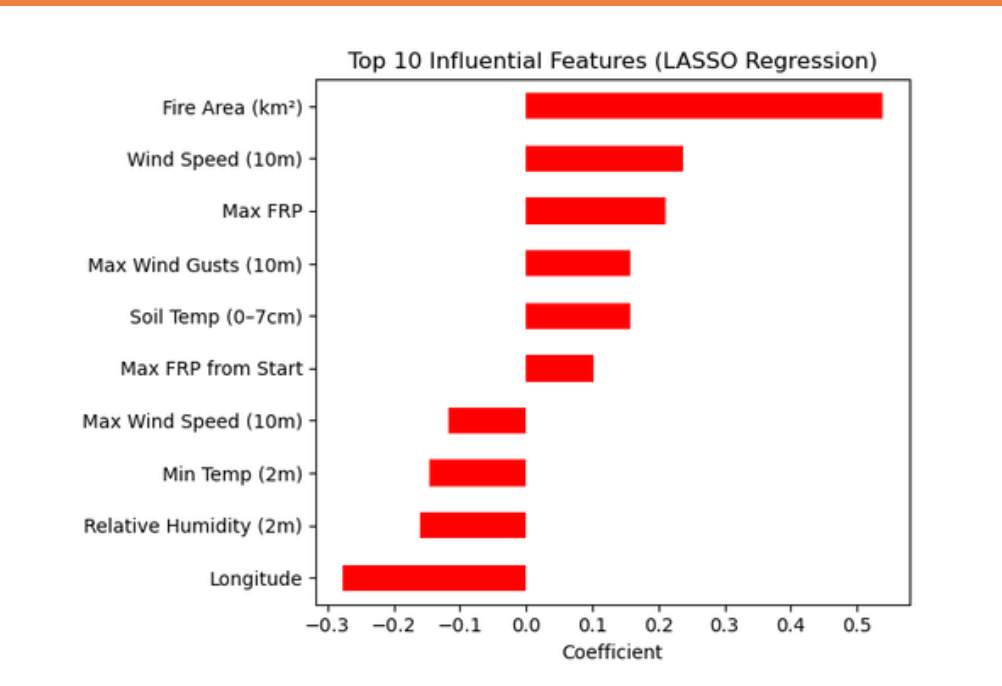
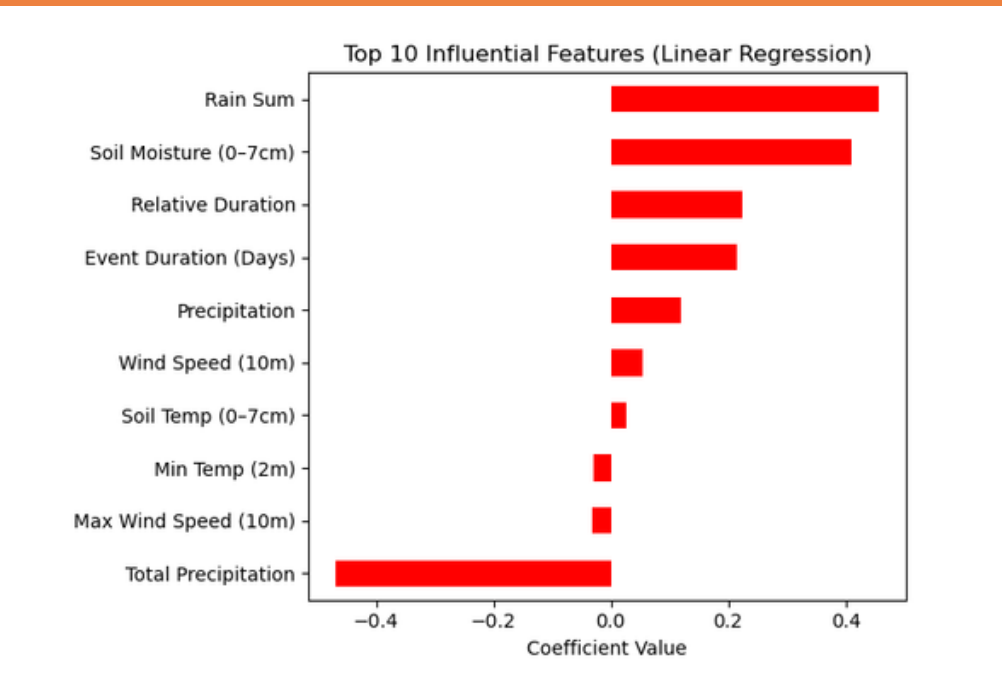


Cleaned fire and smoke data using spatiotemporal joins, then identified unique fire and smoke events through spatial analysis



Important Features Across Models

Smoke Cloud Count (12h) was the top predictor across all models followed by Fire Count and Wind Speed (10m) in LASSO and Random Forest



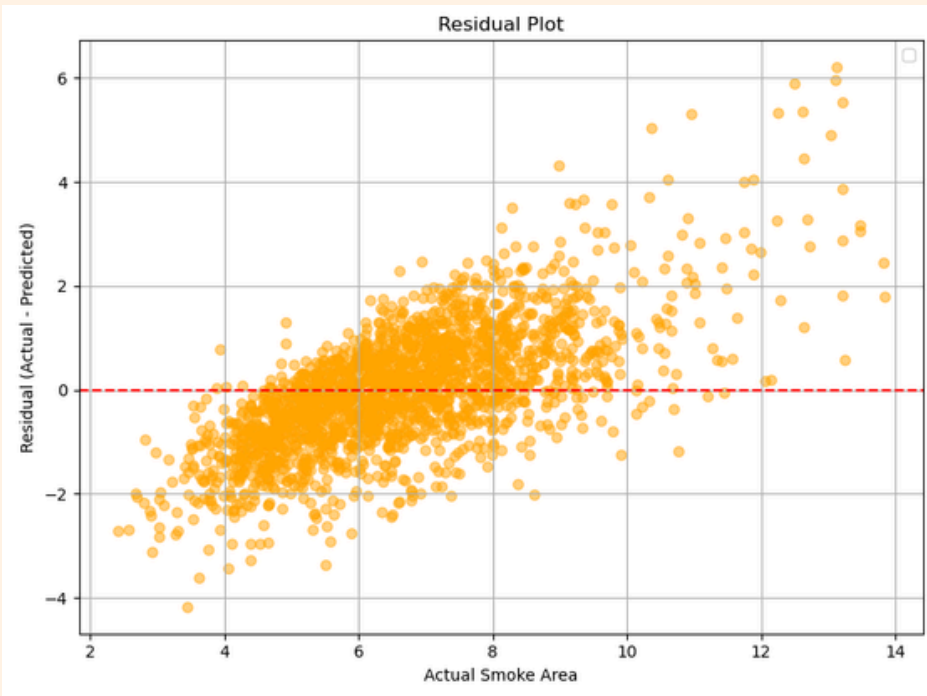
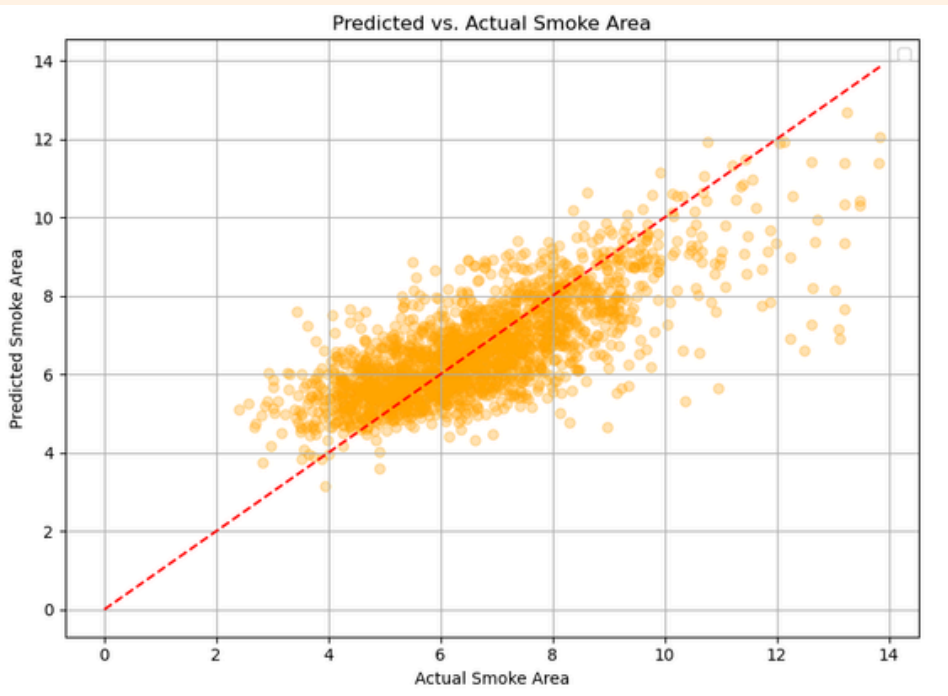
RESULTS

Model Diagnostics:

Method	R ²	RMSE	MAE
Linear Regression	0.201	1.583	1.172
Lasso Regression	0.211	1.577	1.172
Random Forests	0.422	1.344	1.032
XGBoost	0.450	1.311	1.004

Ensemble models perform better than traditional regression, though a low R² indicating that key factors may still be missing.

Final Model Analysis



Notable Features

Fire Area (km²) | Larger fire size generally leads to greater smoke production and wider smoke dispersion.

Wind Speed (10m) | Higher wind speeds can carry smoke particles farther, increasing dispersion and spreading the smoke over a wider area.

Latitude | Acts as a proxy for location factors like wind currents and temperature. The Gulf of Mexico coast, where most fires occur, has a dry biome and vegetation that promote fire activity and smoke spread